

Fall 2026

Math F113X

Math F113X Exam 1

Name: _____

Instructor: _____

Rules:

- Partial credit will be awarded, but you must show your work.
- You may have a $3'' \times 5''$ notecard with writing on both sides.
- Calculators are allowed.
- Turn off anything that might go beep during the exam.

Good luck!

Problem	Possible	Score
1	10	
2	16	
3	20	
4	8	
5	12	
6	15	
7	19	
Extra Credit	(5)	
Total	100	

1. (10 points)

A section of Math F113X with 9 students holds a vote for their favorite vegetable, with the choices being asparagus (A), beets (B) and cabbage (C). The results are as follows:

	Alice	Bob	Cathy	Daniel	Edith	Fahad	Greg	Hyacinth	Ida
1st choice	A	A	A	A	A	B	B	B	C
2nd choice	B	B	C	C	C	A	A	A	A
3rd choice	C	C	B	B	B	C	C	C	B

a. (4 points) Create a preference schedule with this information.

b. (3 points) How many votes are needed for a majority winner? _____
Provide a brief calculation/explanation for your answer.

c. (3 points) What is the smallest number of votes that could give a candidate a plurality? _____
Provide a brief calculation/explanation for your answer.

2. (16 points)

Consider the following preference schedule for an election with 5 candidates and 430 voters.

Rank	120	90	80	67	8	62	3
1st	A	B	C	D	D	E	E
2nd	B	A	B	B	C	B	D
3rd	C	C	A	C	B	D	B
4th	D	D	D	A	A	C	C
5th	E	E	E	E	E	A	A

Answer the following questions. **Show your work.** (This means write down calculations or words so that someone else can understand how you get your answer.)

- a. (4 points) Find the winner using **Plurality voting**. _____
Provide a brief calculation/explanation for your answer.

- b. (2 points) How many votes are needed for a **majority**? _____
Provide a brief calculation/explanation for your answer.

- c. (10 points) Find the winner using **Instant Runoff Voting** (Ranked Choice Voting). _____
Clearly show your work for each round and **state who gets eliminated** in each round.

3. (20 points)

Consider the following preference schedule for an election with 4 candidates and 24 voters.

Rank	6	6	3	9
1st	A	B	A	C
2nd	B	D	D	A
3rd	C	C	C	B
4th	D	A	B	D

- a. (6 points) Find the winner using **Borda Count**. _____
Clearly show your work.

- b. (6 points) Fill in the missing information in the table below and determine the winner using **Copeland's method**.

A vs B	A vs C	A vs D	B vs C	B vs D	C vs D
A:	A: 9	A: 18	B: 12	B: 21	C:
B:	C: 15	D: 6	C: 12	D: 3	D:

Tally: A: _____
B: _____
C: _____
D: _____

Winner under Copeland's method: _____

- c. (4 points) Is there a **Condorcet winner**? If so, who is it? Briefly explain/justify your answer.
- d. (4 points) Did either Borda Count or Copeland's Method violate the **Condorcet criterion** in this situation? If so, which one(s)? Provide a brief explanation of your answer.

4. (8 points)

Choose one of the following methods: **Plurality**, **IRV/RCV**, **Borda Count**, and **Copeland's Method**.

State at least one **strength** your chosen method has over the other three options. State at least one **weakness** your chosen method has.

You should refer to some of the various **fairness conditions** in discussing the weakness of your method, and explain what the fairness condition you are referring to means in the context of your method.

Chosen method: _____

Strength (4 points):

Weakness (4 points):

5. (12 points)

In a certain corporation, shareholders receive 1 vote for each share of stock they hold. A small corporation has two people who have 30 shares each, two people who have 20 shares each, and one person who has only 10 shares.

- a. (3 points) If any decisions require a majority vote, set up a weighted voting system of the form $[q : w_1, \dots, w_n]$ to represent this corporation's shareholder votes.
- b. (3 points) Certain decisions, like amending the bylaws, require a $2/3$ vote. How would you change your weighted voting system to represent this situation?
- c. (3 points) What is the largest value the quota could be for this system? _____
- d. (3 points) What is the smallest value the quota could be for this system? _____

6. (15 points)

Consider the weighted voting system $[18: 8, 5, 4, 3, 3, 2]$. Determine:

- a. (1 point) The total number of players: _____
- b. (1 point) The total weight of the system: _____
- c. (1 point) The quota: _____
- d. (3 points) Are there any dictators? Explain how you know, with a few words.

- e. (3 points) The coalition $\{P_1, P_2, P_3, P_6\}$ is a winning coalition. Explain how you know it is a winning coalition using a computation and some words.

- f. (3 points) In the coalition $\{P_1, P_2, P_3, P_6\}$, is P_6 critical? Explain why or why not, using a computation and some words.

- g. (3 points) Does player P_2 have veto power? Why or why not?

7. (19 points)

Consider the weighted voting system $[27 : 19, 18, 8, 2]$ and the list of all possible coalitions.

coalition	weight	winning?	coalition	weight	winning?
$P_1 P_2$	37		$P_1 P_2 P_3$	45	
$P_1 P_3$	27		$P_1 P_2 P_4$	39	
$P_1 P_4$	21		$P_1 P_3 P_4$	29	
$P_2 P_3$	26		$P_2 P_3 P_4$	28	
$P_2 P_4$	20		$P_1 P_2 P_3 P_4$	47	
$P_3 P_4$	10				

- (2 points) Put a check $\checkmark$ in the column labeled “winning?” to identify all winning coalitions, and mark $\times$ for non-winning coalitions.
- (6 points) In each winning coalition, **underline the critical players**.
- (5 points) Using this information, compute the **Banzhaf Power Index** of each player by filling in the following table.

Player	power

- (2 points) Is there a **dummy**? Why or why not?
- (4 points) Players 1 and 2 have almost the same number of votes, and Player 3 has less than half as many votes as Player 2. Does their decision-making power reflect this vote distribution? Why or why not/how can you tell?

Extra Credit (5 points)

A professional basketball team has five coaches: a head coach (H) and four assistant coaches (A1, A2, A3, and A4).

Constraint: Player personnel decisions require at least three Yes votes, one of which must be from the head coach (H).

If h is the weight assigned to the head coach H , and a is the weight assigned to each of the four assistant coaches (assuming they each get the same number of votes), determine values

$$[q : h, a, a, a, a]$$

that describe the weighted voting system. Explain why your assignment of values matches the constraint given above.